

Monitoring in crop protection

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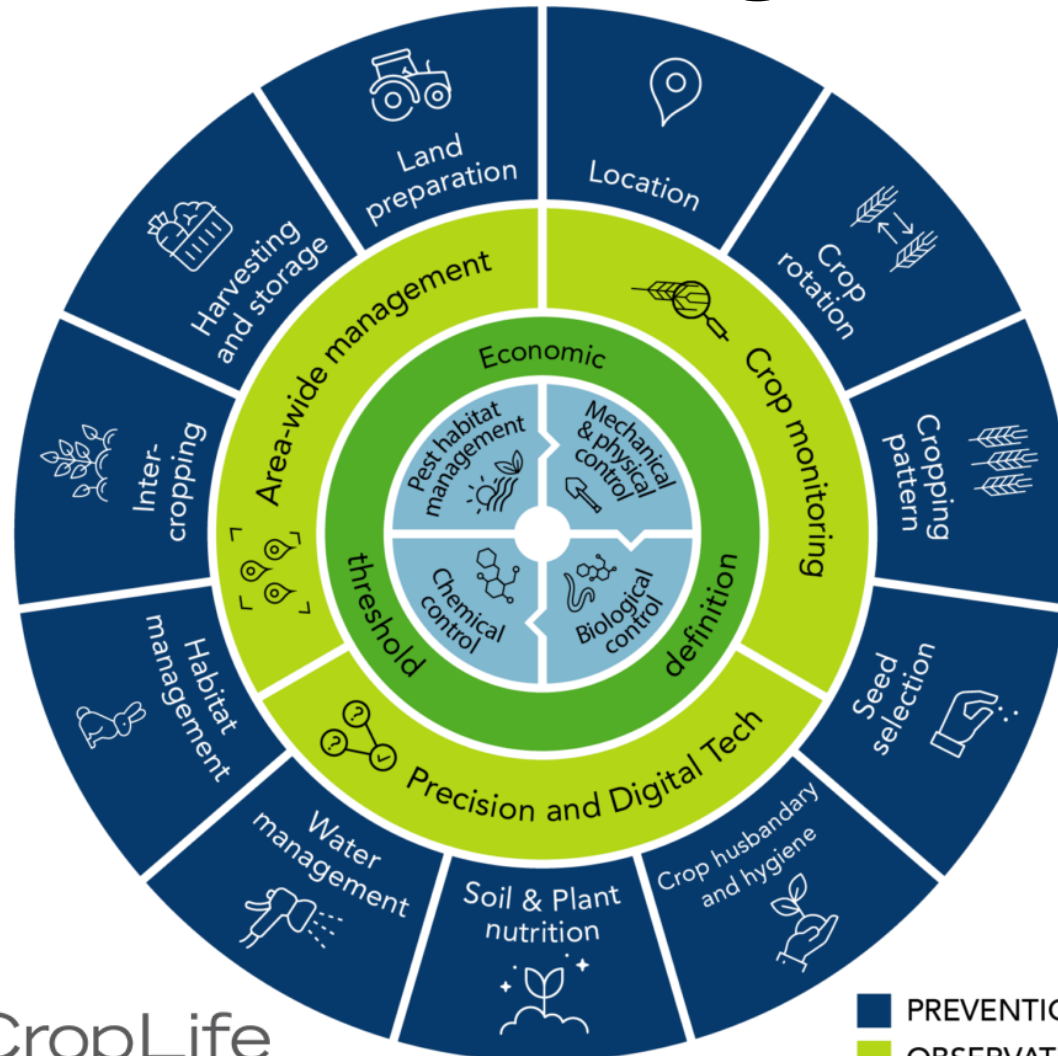
A working definition of Integrated Pest Management (IPM)

Kogan *et al.* synthesized these points into the following definition:

“IPM is a decision support system for the selection and use of pest control tactics, singly or harmoniously coordinated into a management strategy, **based on cost/benefit analyses** that take into account the interests of and impacts on producers, society, and the environment”

Kogan (1998) Annual Review of Entomology. 43: 243-270

Integrated Pest Management (IPM)



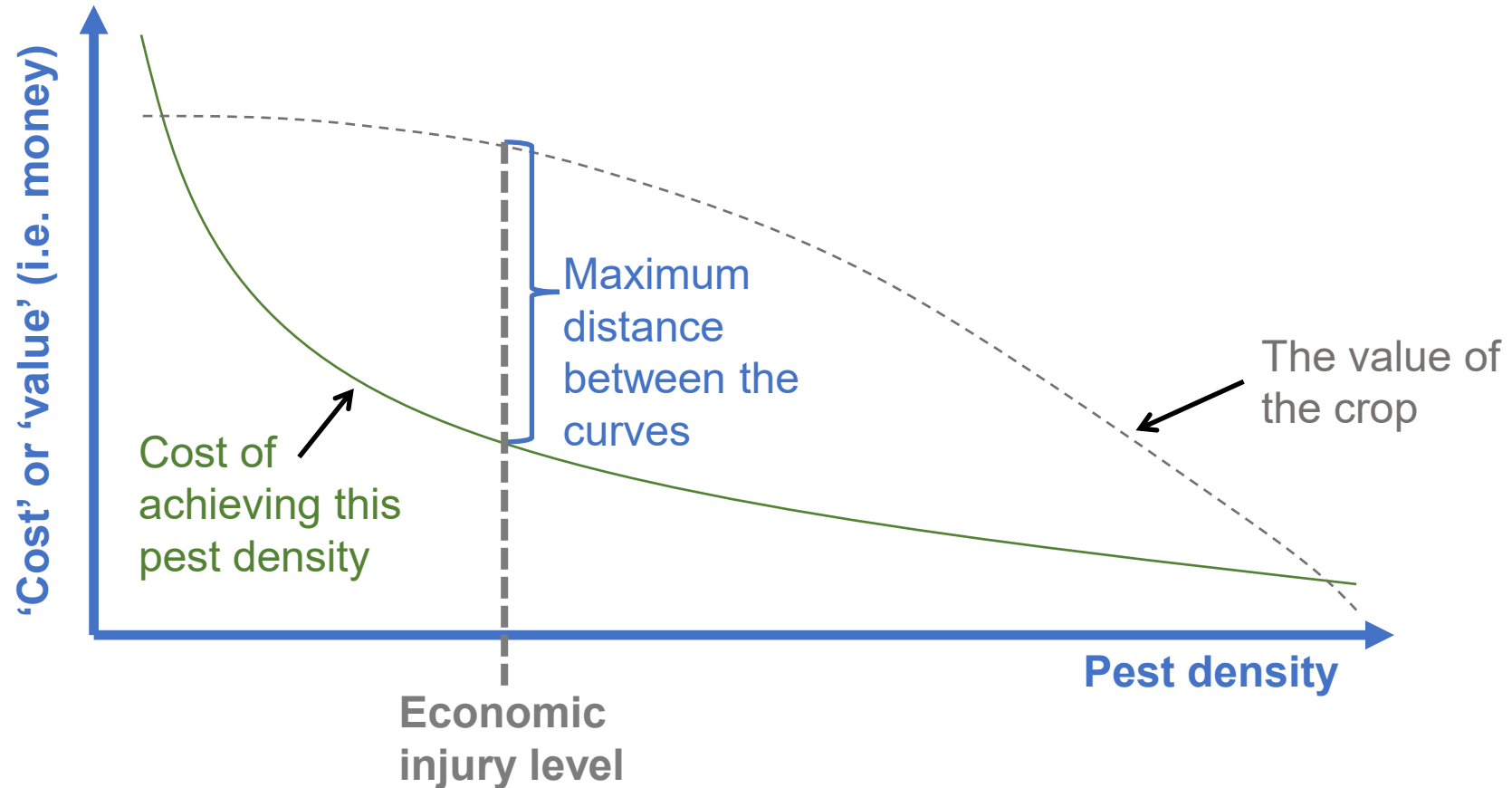
Economic injury

- Do we need to eradicate pests?
- What level of damage can we tolerate?
- When do we need to control a pest?

Economic injury (EIL) is:

‘the pest population below which the extra cost of control exceeds any additional revenue from improved yield or quality’

Economic injury level



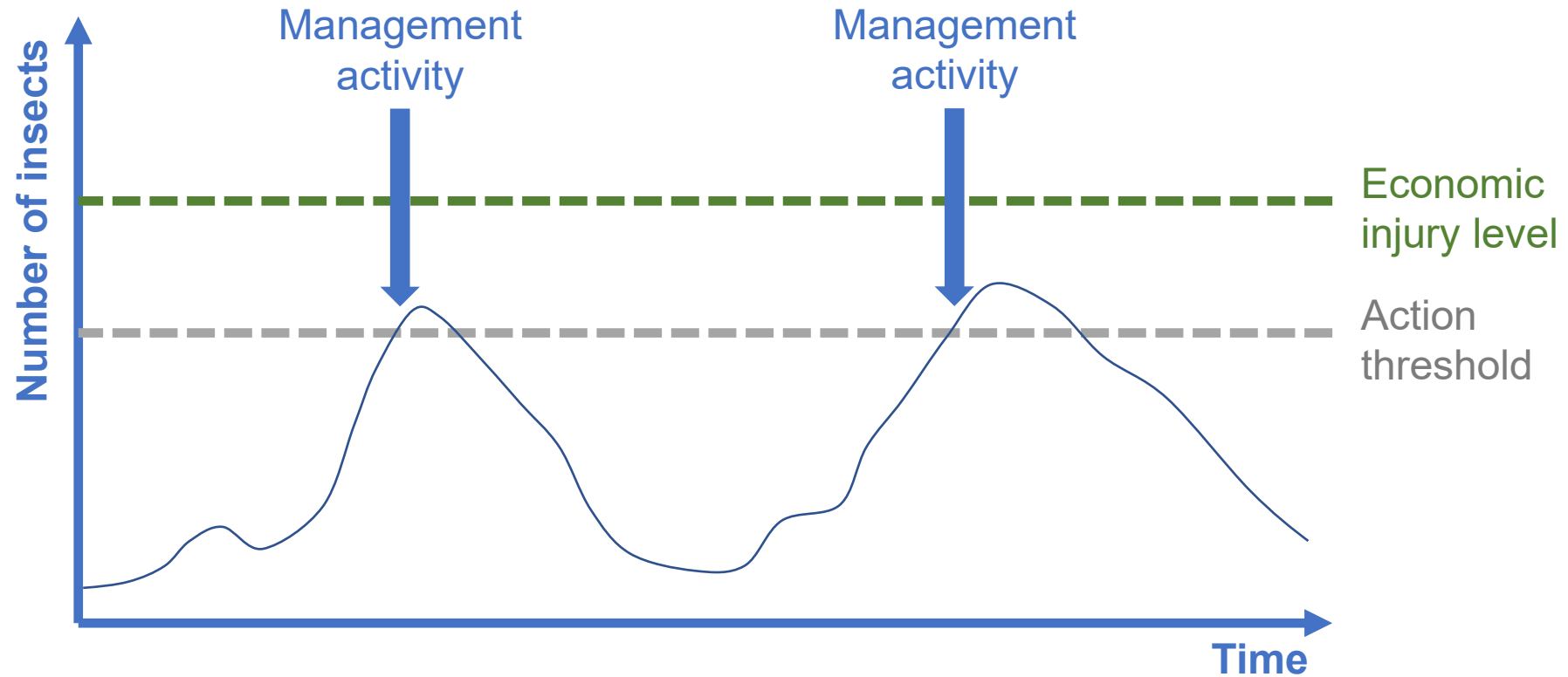
Useful, but a theoretical value that is difficult to use in practice

Economic threshold or action threshold (Pt1)

‘a current pest density that represents a future population, the control of which will prevent economic loss equal to the cost of implementing the control tactic’

The time to take action!

Economic threshold or action threshold (Pt2)



How do we monitor crop pests? (Pt1)

- Count pests on the plant:
 - Simple, requires no equipment
 - Can give a rapid answer (depends on pest)

But

- May be time consuming (again depends on pest)
- May require detailed knowledge of the pest
- Only a small area of the crop sampled
- May need to be repeated



How do we monitor crop pests? (Pt2)

- Count pests in soil samples:
 - May provide advanced warning
 - May complement other monitoring techniques

But

- Soil samples often need to be processed in a laboratory
- Time consuming and expensive
- Not always reliable
- Only appropriate for pests which complete at least part of their lifecycle in the soil



How do we monitor crop pests? (Pt3)

- Trap pests using artificial refuges:



How do we monitor crop pests? (Pt4)

- Count pests within refuge:
 - May provide advanced warning
 - Useful for nocturnal pests or pests that may spend time in the soil

But

- Refuges may need to be checked regularly
- Catches may be determined by other factors e.g. natural refuges, food sources and environmental conditions
- Area sampled depends on mobility of species

How do we monitor crop pests? (Pt5)

- Record pest damage:



How do we monitor crop pests? (Pt6)

- Record pest damage:
 - Avoids the need to find the pest itself
 - Damage is often more obvious and can be quickly recorded

But

- Recording pest damage may not always be pest specific
- It can be difficult to quantify pest damage i.e. often presence/absence
- Too late to prevent damage, this may not be acceptable in certain crops

How do we monitor crop pests? (Pt7)

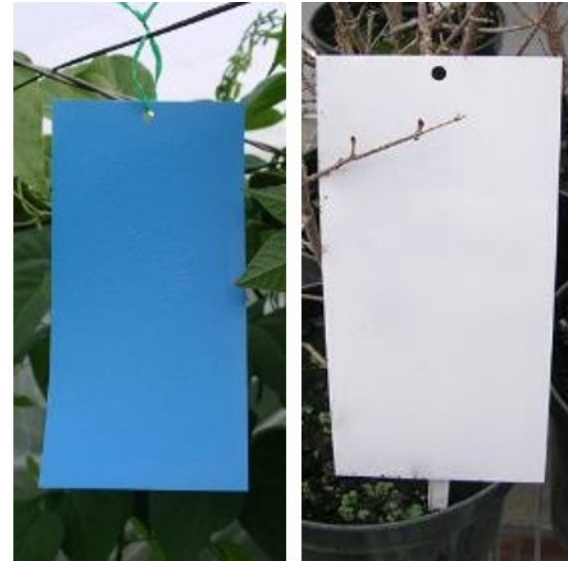
- Trap pest using colour:



Yellow sticky
trap



Yellow water
trap



Other colours of
sticky trap

How do we monitor crop pests? (Pt8)

- Trap pests using colour:

- Quick and easy to use
- Usually effective

But

- Not species specific, may require specialist knowledge
- Traps need to be checked regularly



How do we monitor crop pests? (Pt9)

- Trap pests using odour (e.g. pheromones):



Typical pheromone trap
(Delta trap)



Other pheromone trap
designs e.g. cross vein and
funnel



How do we monitor crop pests? (Pt10)

- Trap pests using odour (e.g. pheromones):
 - Very effective, traps pest even at low densities
 - Species specific

But

- Usually only traps males
- Traps need to be checked regularly
- Traps need to be calibrated
- May be expensive



Rubber septum
impregnated with
pheromone

How do we monitor crop pests? (Pt11)

- Smart trap using machine vision and drones are increasingly being used:



‘Magic traps’ used for cabbage stem flea beetle monitoring



‘RapidAim’ codling moth monitoring



‘SkipPyScout’ rapid crop walking – not machine vision

How do we monitor crop pests? (Pt12)

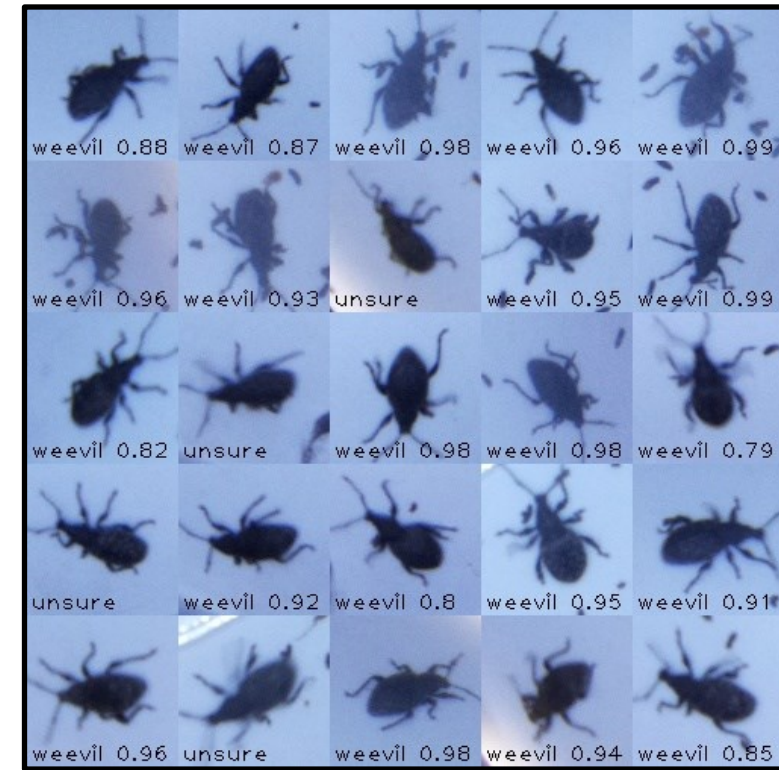
- Smart trap using machine vision are increasingly being used:
 - Reduces time taken to monitor for specific species
 - Species identification completed by the machine not the human

But

- Only developed for some species
- Currently each unit is very expensive
- May be issues/costs associated with mobile communication

How do we monitor crop pests? (Pt13)

- Developing a smart trap for a nocturnal pest species:



How do we monitor crop pests? (Pt14)

- National monitoring schemes:
 - University of Warwick – [Pest Activity](#)
 - Brassica pests e.g. cabbage root fly, aphids, pollen beetle
 - Carrot & parsnip pests e.g. carrot fly, aphids, cutworm
 - Leaf & onion e.g. thrips, bean seed fly, cutworm



Cabbage root fly larvae



Cutworm



Thrips damage on onion

How do we monitor crop pests? (Pt15)

- National monitoring schemes:
 - Rothamsted Insect Survey: [Rothamsted Insect Survey](#)
 - Aphid suction trap network
 - Light trap network



Rothamsted
suction trap



Rothamsted
light trap

How do we monitor crop pests? (Pt16)

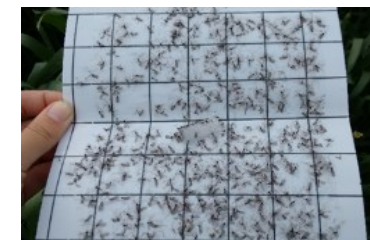
Aphid Bulletin 11:

09 06 2025 - 15 06 2025

	Iv	D	G	Ay	N	Y	K	HA	BB	We	H	RT	Wr	SP	EM	SX
Days	3	6			7	6	3	5	7	7	3	7	7	5	7	6
Part Catches	0	2			0	0	0	0	0	0	0	0	0	0	0	0
<i>Acyrtosiphon pisum</i>			2		3	2	5	26	72	110	6	53	97	8	52	
<i>Aphis fabae</i>			2		3	2	2	8	328	18	4	98	116	46	330	24
<i>Aulacorthum solani</i>	2	1			1		1		10	11		2			3	5
<i>Brachycaudus helichrysi</i>	5	21			16	52	16	32	102	71	39	52	76	13	101	5
<i>Brevicoryne brassicae</i>	9	5			2	44	15	332	172	243	55	69	133	58	122	13
<i>Cavariella aegopodii</i>	5	2			10	6	8	20	40	59	42	27	37	14	24	8
<i>Drepanosiphum platanoidis</i>	4	35			205	114	18	37	16	17	13	56	60	24	12	18
<i>Elatobium abietinum</i>	3	9			9			1								3
<i>Hyalopterus pruni</i>	1	4			8	10	10	8	28	21	10	33	39	33	63	10
<i>Hyperomyzus lactucae</i>			3			6	6	11	16	36	3	9	20	2	26	
<i>Macrosiphum euphorbiae</i>	1	8			7	26	2	78	40	46	51	17	34	10	23	3
<i>Metopolophium dirhodum</i>	9	23			23	20	40	63	116	43	25	47	65	10	41	10
<i>Myzus ascalonicus</i>			3		8	18	2	7	4	1	1	1			2	
<i>Myzus persicae</i>	21	9			17	96	27	150	474	144	48	133	217	8	144	118
<i>Nasonovia ribisnigri</i>	1					2				2	1			1		2
<i>Phorodon humuli</i>			1		2	110	34	30	176	65	58	83	357	65	226	17
<i>Rhopalosiphum maidis</i>										1			4		2	

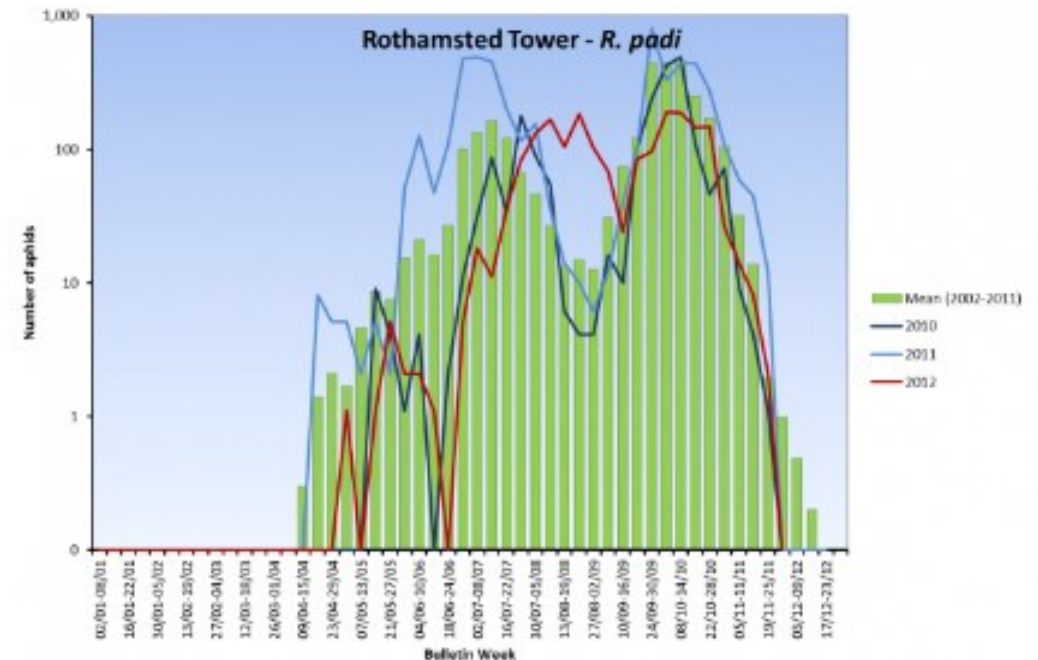
How do we assess pests? (Pt1)

- Timing – should allow time for management decisions to be taken
- Meaningful – can we use knowledge of pest levels and crop growth stage in management decisions
- Proportionate – effort taken to assess pest levels should not exceed value of information



How do we forecast pests? (Pt1)

- Pest forecasts require an understanding of the pests biology
- Often forecasts are
 - Based on historical records
 - Based on meteorological



Long-term trends in numbers of migrating bird cherry-oat aphids

How do we forecast pests? (Pt2)

- Aphid forecasts
 - Based on average temperatures in January and February
 - Colder temperatures in January and February = fewer aphids in the spring and aphids fly to cereal crops later



Bird cherry-oat aphid

How do we forecast pests? (Pt3)

- Aphid forecasts – bird cherry-oat aphid

		1st capture in suction trap			Numbers to 1st July		
		Predicted	Prediction date range (75% Confidence limits)	Ranking	Predicted	75% Confidence limits	Ranking
Suction-trap sites:	Mean Jan-Feb						
Dundee (D)	4.07	17 May	(26 April - 7 June)	=23/56	41	(4 - 361)	22/54
Gogarbank (G)	4.62	11 May	(21 April - 1 June)	=20/57	56	(11 - 266)	26/57
Ayr (Ay)	4.82	unavailable					
Newcastle (N)	4.84	18 May	(28 April - 8 June)	=23/57	28	(5 - 155)	=24/56
York (Y)*	2.91	28 April	(7 April - 18 May)	7/11	52	(8 - 314)	10/11
Kirton II (K)	4.10	25 April	(31 March - 19 May)	=22/46	111	(24 - 494)	26/46
Broom's Barn (BB)	4.37	28 April	(10 April - 17 May)	23/58	134	(29 - 614)	28/57
Wellesbourne (We)	4.29	23 April	(23 March - 24 May)	14/21	71	(32 - 159)	14/18
Hereford (H)	unavailable	21 April	(5 March - 7 June)	=19/53	51	(8 - 298)	27/52
Rothamsted (RT)	3.81	4 May	(13 April - 25 May)	=28/58	58	(13 - 249)	30/58
Writtle (Wr)	4.14	29 April	(30 March - 29 May)	29/51	88	(19 - 393)	34/51
Silwood (SP)	3.89	29 April	(31 March - 27 May)	=26/47	63	(16 - 237)	26/43
Starcross (SX)	5.38	28 March	(14 February - 10 May)	=22/56	218	(53 - 895)	27/54

Summary

- Monitoring insects is central to decision making within Integrated Pest Management
- Tools used are often very similar to those employed in ecological surveys
- New technologies are helping to improve species monitoring
- However, there is still a way to go to improve how we use monitoring information to inform use of action thresholds